# FIELDCOMMS

## EmComm Field Server v1.0

### INSTALLATION GUIDE

Raspberry Pi 5 (16 GB) · Pironman MAX 5 · 2× 1 TB NVMe RAID 1

Offline Emergency Communications Server

Version 1.0 · June 21, 2026

McHenry County Emergency Services Volunteers and McHenry County Emergency Management Agency · K9ESV

| | | | | |
|---|---|---|---|---|
| **29 Web Pages** | **10 Background Services** | **Kiwix Offline Library** | **3 Dashboard Modes** | **APRS · Winlink · ICS · JS8Call** |

# Table of Contents


| | | |
|---|---|---|
| 1. | Overview & What's Included | 3 |
| 2. | Hardware Requirements | 4 |
| 3. | Before You Begin — Prerequisites | 6 |
| 4. | Step 1: Network Hardware Setup (ASUS Router + UniFi Switch) | 7 |
| 5. | Step 2: Download & Run the FieldComms Installer | 9 |
| 6. | Step 3: Installer Configuration Options | 11 |
| 7. | Step 4: Pi Static IP Configuration | 12 |
| 8. | Step 5: Kiwix Offline Library Setup | 13 |
| 9. | Step 6: FCC Amateur Database | 14 |
| 10. | Step 7: APRS Setup (Graywolf + YAAC) | 15 |
| 11. | Step 8: Pat Winlink — Verify & Configure | 16 |
| 12. | Step 9: Windows Laptop (Winlink Express + JS8Call) | 17 |
| 13. | Step 10: First Boot Verification | 19 |
| 14. | Network Architecture Reference | 20 |
| 15. | Web Dashboard Reference | 21 |
| 16. | Service & Port Reference | 23 |
| 17. | Maintenance & Updates | 24 |
| 18. | Troubleshooting | 25 |
| 19. | Quick Reference Card | 27 |

# 1. Overview & What's Included

FieldComms is a complete off-grid emergency communications server for Raspberry Pi. It provides a WiFi access point (EMCOMM-NET) that any phone, tablet, or laptop on scene can connect to, then access all EmComm tools through a web browser — no internet required. When Ethernet internet is available, live features such as NWS weather alerts, HF propagation data, and APRS-IS automatically activate.

## What's Included

| COMPONENT | DESCRIPTION | COMPONENT | DESCRIPTION |
|---|---|---|---|
| 26 Web Pages | Full-featured dashboard, net logs, ICS forms, maps, roster, cheat sheets | Dead Man's Switch | Per-net inactivity monitor with warning and trigger states |
| Net Control Logger | Amateur + Starcom nets, FCC autofill, ICS-309 export | Pre-Flight Checklist | GO/CAUTION/NO-GO deployment readiness check |
| ICS Platform | Command, Operations, Planning, Logistics, Finance sections | HF Propagation | Solar indices, band conditions, MUF/LUF, ionosphere diagram |
| Tactical APRS Map | Leaflet map — Graywolf + YAAC dual-source, live WebSocket | Repeater Database | RepeaterBook offline import, filter, ARES/SKYWARN badges |
| Member Roster | 11 certifications, 13 equipment fields, activations, CSV import | Facilities Directory | EOC, hospital, shelter CRUD with radio frequencies |
| NTS Radiogram | ARRL-format radiogram generator and message log | Grid Square Calculator | Maidenhead bidirectional, distance/bearing, canvas map |
| Resource Board | Personnel, vehicles, equipment status tracking | Kiwix Library | Offline WikiMed, Wikipedia, iFixit, Wikibooks (port 8081) |
| FCC Callsign Lookup | Local SQLite DB — full amateur license database, offline | Print Center | Unified print hub — ICS forms, radiograms, cover sheets |
| Radio Cheat Sheets | Phonetic, Q-codes, prowords, band plan, CTCSS, RST | Health Monitor | CPU, disk, GPS, all services — port 5051 |

# 2. Hardware Requirements

## Supported Hardware — Pi Server

FieldComms v1.0 runs on the **Raspberry Pi 5 with 16 GB RAM**. The 16 GB model provides ample processing power and memory for all FieldComms services, Kiwix, APRS, and ICS simultaneously.

| CONFIGURATION | STORAGE | NOTES |
|---|---|---|
| Pi 5 (16 GB) + Pironman MAX 5 enclosure | 2× 1 TB NVMe SSD RAID 1 mirror | Tower form factor with OLED display, active cooling, dual M.2 NVMe slots. Two 1 TB SSDs configured as RAID 1 — drives mirror each other for data redundancy. If one SSD fails, the system keeps running on the other with no data loss. |

## Network Hardware

Wi-Fi is provided by the ASUS RT-BE58 Go travel router — the Pi does not run a Wi-Fi hotspot. The UniFi switch provides wired 2.5 GbE ports for the Pi and other fixed devices.

| DEVICE | ROLE | NOTES |
|---|---|---|
| ASUS RT-BE58 Go | Wi-Fi 7 access point + DHCP server | Dual-band Wi-Fi 7 (802.11be), 2.5G LAN port, USB-C 18W power. Broadcasts EMCOMM-NET on 2.4 GHz and 5 GHz. Powered by USB-C power bank for field use. |
| UniFi Switch Flex 2.5G-5 (USW-Flex-2.5G-5) | 5-port 2.5 GbE wired switch | 4× 2.5 GbE ports + 1× 2.5 GbE PoE input. Powered via USB-C (5V/1A) or PoE. Port 1 → ASUS router, Port 2 → Pi, Ports 3–5 spare. |
| Windows Laptop | Winlink Express + JS8Call station | Connected to EMCOMM-NET (Wi-Fi) or UniFi switch (wired). Runs IC-7300 via single USB cable. Separate from Pi. |

## Accessories & Cables

| ITEM | REQUIRED | PURPOSE |
|---|---|---|
| 2× 1 TB M.2 NVMe SSD (for Pironman MAX 5 dual slots) | Required | Configured as RAID 1 (mirror) — OS, FieldComms, FCC DB, Kiwix ZIMs, logs. Mirroring provides automatic redundancy: one drive can fail without data loss. |
| Pi 5 USB-C power supply (27W PD) | Required | Official Raspberry Pi 27W USB-C PD supply or equivalent |
| ASUS RT-BE58 Go travel router | Required | Wi-Fi 7 access point — provides EMCOMM-NET for all field devices |
| UniFi Switch Flex 2.5G-5 | Required | 5-port 2.5 GbE switch — distributes wired connections at the EOC |
| Cat 6 Ethernet cables (×3 minimum) | Required | Router → Switch, Switch → Pi, Switch → spare port |
| USB-C cable + power bank (≥20,000 mAh, 18W+ PD) | Recommended | Field power for ASUS router and UniFi switch — 6–8 hrs runtime |

| | | |
|---|---|---|
| USB-A drive (32 GB+, labeled FIELDCOMMS) | **Recommended** | Auto-backup trigger — insert to back up all runtime data instantly |
| External USB hard drive — 1 TB+ (e.g. LaCie Rugged or LaCie Mobile Drive) | **Recommended** | Incident archive and full system backup. LaCie Rugged is shock/drop/crush resistant — ideal for field use. Label FIELDCOMMS for auto-backup compatibility. |
| USB GPS receiver (u-blox or GlobalSat) | Optional | Live position for tactical map and NWS alerts |
| USB TNC — Digirig Mobile or SignaLink USB | Optional | Required for APRS transmit/receive via Graywolf or YAAC. FieldComms assigns a stable /dev/tnc0 device name via udev. |
| Powered USB hub (4- or 7-port, with power supply) | Optional | Recommended if connecting GPS + TNC + backup drive simultaneously. Must be a powered hub — unpowered hubs can cause instability. |
| USB Printer (e.g. Brother HL-L2350DW, HP LaserJet) | Optional | Shared across EMCOMM-NET via CUPS (installed automatically). Or connect a Wi-Fi printer directly to EMCOMM-NET. Battery printers: Canon PIXMA TR150, HP OfficeJet 200. |

## SD Card / Storage Sizing Guide

| CARD SIZE | WHAT FITS | RECOMMENDED FOR |
|---|---|---|
| 16 GB | OS + FieldComms + FCC DB + Kiwix Tier 1 (WikiMed, Wikipedia Mini, Wikivoyage) | Minimal field deployment, basic reference |
| 32 GB | Above + Kiwix Tier 2 (Wikibooks, iFixit) + APRS data | Standard field server — recommended minimum |
| 64 GB | Above + Kiwix Tier 3 (Full Medical Wiki, Wikiversity, Electronics SE) | Full EOC deployment, extended operations |
| 2× 1 TB NVMe (RAID 1) | All ZIM tiers + full logs, backups, incident archives, multiple Kiwix updates — ~900 GB usable after RAID 1 mirroring overhead | K9ESV standard deployment — Pironman MAX 5 with dual NVMe RAID 1 |

## USB Hub & Multi-Device Setup

The Pi 5 has four USB ports (2× USB 3.0, 2× USB 2.0). A powered USB hub lets you connect a GPS receiver, TNC (Digirig, SignaLink, etc.), LaCie backup drive, and other peripherals simultaneously without running out of ports. FieldComms uses **udev rules** to assign stable device names — so the GPS always appears as **/dev/gps0** and the TNC always appears as **/dev/tnc0**, regardless of which hub port they are plugged into or the order they are powered on.

**Recommended:** use a **powered** USB hub (with its own power supply). Unpowered hubs draw power from the Pi and can cause instability with multiple devices connected.

| DEVICE | STABLE NAME | HOW IT IS ASSIGNED | SERVICE THAT USES IT |
|---|---|---|---|
| USB GPS receiver (GlobalSat BU-353, u-blox, etc.) | `/dev/gps0` | udev rule matches by USB vendor/product ID. Stable regardless of hub port or plug order. | gpsd → tactical map + NWS alerts |

| | | | |
|---|---|---|---|
| Digirig Mobile TNC | `/dev/tnc0` `/dev/digirig` | udev rule matches by vendor ID + product string "Digirig Mobile". Distinguished from GPS even though both use CP2102 chip. | YAAC / Graywolf APRS |
| SignaLink USB TNC | `/dev/tnc0 /dev/signalink` | udev rule matches by Texas Instruments PCM2904 USB audio chip. No conflict with GPS — completely different vendor ID. | YAAC / Graywolf APRS |
| LaCie / USB backup drive (labeled FIELDCOMMS) | `/dev/sdX (auto-mounted)` | udev backup rule triggers on label FIELDCOMMS. Drive letter (sda, sdb) is assigned dynamically — label is used, not path. | fieldcomms-backup@.service (auto-triggered) |

## The Digirig / GPS Conflict — Why It Matters

Both the **Digirig Mobile** TNC and the **GlobalSat BU-353-S4** GPS use the same USB chip (Silicon Labs CP2102, vendor ID 10c4, product ID ea60). Without careful rules, Linux would assign them both to the same device path or give them to the wrong service. FieldComms solves this by matching on the USB product description string in addition to the vendor/product IDs:

```
# See what product string your device reports:
udevadm info -a -n /dev/ttyUSB0 | grep -E "idVendor|idProduct|product"

# Digirig Mobile reports:
#   ATTRS{idVendor}=="10c4"
#   ATTRS{idProduct}=="ea60"
#   ATTRS{product}=="Digirig Mobile"     ← used to separate from GPS

# GlobalSat BU-353-S4 reports:
#   ATTRS{idVendor}=="10c4"
#   ATTRS{idProduct}=="ea60"
#   ATTRS{product}=="CP2102 USB to UART Bridge Controller"  ← different!

# GPS rule excludes Digirig:
# ATTRS{product}!="Digirig Mobile"  → only GPS gets /dev/gps0

# TNC rule requires Digirig product string:
# ATTRS{product}=="Digirig Mobile"  → only Digirig gets /dev/tnc0
```

## Verifying Device Assignment

After plugging in your devices, run these commands to confirm the stable symlinks are working correctly:

```
# List all USB serial devices
ls -la /dev/ttyUSB* /dev/ttyACM* 2>/dev/null

# Check the stable symlinks
ls -la /dev/gps0 /dev/tnc0 2>/dev/null

# Should show something like:
# lrwxrwxrwx 1 root root 7 Jun 21 /dev/gps0 -> ttyUSB0
# lrwxrwxrwx 1 root root 7 Jun 21 /dev/tnc0 -> ttyUSB1

# Confirm GPS is outputting NMEA sentences
sudo cat /dev/gps0
# Should show: $GPRMC,123519,A,4807.038,N,01131.000,E,...

# Confirm gpsd can see the GPS
gpspipe -w -n 5

# Check what device YAAC should use for the TNC
ls -la /dev/tnc0
# Use /dev/tnc0 in YAAC → Configure → Ports → Serial TNC
```

### YAAC TNC Port Configuration

When configuring YAAC to use your TNC, use the stable symlink name **/dev/tnc0** rather than **/dev/ttyUSB0**. This ensures YAAC connects to the correct device even if the hub port order changes or a device is replugged.

| TNC TYPE | YAAC PORT TYPE | DEVICE PATH | BAUD RATE |
|---|---|---|---|
| Digirig Mobile | `Serial TNC (KISS)` | `/dev/tnc0` | `9600` |
| SignaLink USB | `Serial TNC (KISS)` | `/dev/tnc0` | `9600` |
| FTDI-based TNC (RigBlaster, BayPac, etc.) | `Serial TNC (KISS)` | `/dev/tnc_ftdi` | `9600 or 1200` |
| Direwolf software TNC (sound card) | `KISS TNC (TCP)` | `localhost:8001` | `N/A (TCP)` |

- If your TNC does not appear at /dev/tnc0 after plugging in, run: udevadm info -a -n /dev/ttyUSB0 | grep -E "idVendor|idProduct|product" and compare the product string to the rules in /etc/udev/rules.d/99-fieldcomms-tnc.rules. Add a new rule matching your device's exact product string if needed, then run: sudo udevadm control --reload-rules && sudo udevadm trigger

## Complete Bill of Materials — K9ESV Standard Deployment

The following is the complete equipment list for building the standard MCESV/MCEMA FieldComms deployment. All items are available from Amazon, the Raspberry Pi Foundation, and major electronics retailers. Check current prices at the sources listed — component costs vary.

| ITEM | MODEL / SPEC | WHERE TO BUY |
|---|---|---|

| | | |
|---|---|---|
| **— Core Server —** | | |
| Raspberry Pi 5 — 16 GB RAM | Raspberry Pi 5 Model B (16 GB) | raspberrypi.com · Adafruit · PiShop.us |
| Pironman MAX 5 enclosure | Pironman MAX 5 (tower, dual NVMe, OLED, fans) | 52pi.com · Amazon |
| NVMe SSD × 2 (for RAID 1) | 1 TB M.2 2280 PCIe Gen 3/4 NVMe (e.g. WD Blue SN580) | Amazon · B&H; · Newegg |
| Pi 5 USB-C power supply | Official Raspberry Pi 27W USB-C PD power supply | raspberrypi.com Adafruit · Amazon |
| MicroSD card (boot only / initial RAID setup) | 32 GB Class 10 / A1 microSD (any brand) | Amazon |
| **— Networking —** | | |
| ASUS RT-BE58 Go travel router | ASUS RT-BE58 Go (Wi-Fi 7, 2.5G LAN, USB-C power) | Amazon · Best Buy · B&H; |
| UniFi Switch Flex 2.5G-5 | Ubiquiti USW-Flex-2.5G-5 (5-port 2.5 GbE) | ui.com · Amazon · B&H; |
| CAT 6 Ethernet cables × 4 | 3 ft – 10 ft CAT 6 patch cables (router→switch, switch→Pi, spares) | Amazon · Monoprice |
| **— Radio & Comms —** | | |
| Icom IC-7300 HF transceiver | Icom IC-7300 (HF/50 MHz, built-in USB sound card) | Ham Radio Outlet DX Engineering Amazon |
| USB-A to USB-B cable (IC-7300 to laptop) | USB-A to USB-B, shielded, 6 ft — single cable for audio + CAT + PTT | Amazon · Monoprice |
| Windows laptop (Winlink Express + JS8Call) | Any Windows 10/11 laptop with USB-A port and Wi-Fi | Existing equipment Best Buy · Amazon |
| **— Accessories —** | | |
| Power bank for router + switch (field power) | 20,000+ mAh, 18W+ USB-C PD output (e.g. Anker 737, Baseus 65W) | Amazon |
| USB flash drive (labeled FIELDCOMMS) | 32 GB+ USB 3.0 drive — auto-backup trigger when inserted | Amazon |
| External USB hard drive — incident archive & full backup (e.g. LaCie Rugged or LaCie Mobile Drive) | 1 TB+ USB 3.0/USB-C portable hard drive LaCie Rugged recommended — shock/drop/crush resistant, ideal for field use Label drive FIELDCOMMS for auto-backup trigger compatibility | B&H; · Amazon Best Buy |
| USB GPS receiver (optional) | GlobalSat BU-353-S4 or u-blox USB GPS puck | Amazon |
| USB TNC — Digirig Mobile (optional) | Digirig Mobile v1.x — USB soundcard + serial CAT control in one unit Works with Winlink Express, JS8Call, Vara, and APRS via YAAC/Graywolf Requires /dev/tnc0 udev rule (installed automatically) | digirig.net Amazon |

| | | |
|---|---|---|
| USB TNC — SignaLink USB (optional) | Tigertronics SignaLink USB — USB soundcard interface Alternative to Digirig — plug-and-play with most rigs Rig-specific cable required (order with the unit) | tigertronics.com Ham Radio Outlet Amazon |
| Powered USB hub (optional — recommended if using GPS + TNC + backup drive) | 4-port or 7-port powered USB 3.0 hub with its own power supply Powered hub required — unpowered hubs can cause instability Recommended: Anker 7-Port USB 3.0 Data Hub with power adapter | Amazon Best Buy |
| USB Printer — laser (recommended) | Brother HL-L2350DW — laser, USB + Wi-Fi, great Linux support HP LaserJet P1102w — laser, USB + Wi-Fi, HPLIP driver Shared via CUPS after plugging into Pi or USB hub | Best Buy Amazon Ham Radio Outlet |
| USB Printer — portable/battery (field option) | Canon PIXMA TR150 — battery-powered, USB + Wi-Fi, ~200 pages per charge HP OfficeJet 200 — battery-powered, larger paper tray For activations without shore power | Best Buy Amazon Office Depot |
| Avery 5371 business card sheets (operator cards) | Avery 5371, 10 cards per sheet — laser or inkjet | Office Depot Amazon · Staples |

■ The IC-7300 and Windows laptop are listed separately as they are typically existing club or personal equipment. Check current pricing at Ham Radio Outlet, DX Engineering, or your preferred ham radio dealer.

## AiMesh Wi-Fi Extension Kit — Large Venue Deployments

For deployments where a single ASUS RT-BE58 Go does not cover the entire operational area, add one or more AiMesh nodes. See Section 1.5 for setup instructions. The following equipment is needed per additional node:

| ITEM | MODEL / SPEC | WHERE TO BUY |
|---|---|---|
| AiMesh node router (wireless backhaul) | ASUS RT-BE58 Go (same as primary — simplest pairing) *or any AiMesh-compatible ASUS router* | Amazon Best Buy · B&H; |
| AiMesh node router (high-capacity venues) | ASUS ZenWiFi Pro ET12 or RT-AX88U Pro *Better range and capacity for large buildings* | Amazon B&H; · Costco |
| CAT 6 Ethernet cable (wired backhaul — recommended) | CAT 6 cable from UniFi switch to node LAN port Length as needed for your site | Amazon Monoprice Home Depot |
| USB-C power bank (field power for node) | 10,000+ mAh, 18W+ USB-C PD output One per node for field deployment without shore power | Amazon |

| DEPLOYMENT SCENARIO | RECOMMENDED SETUP |
|---|---|
| Single room EOC (≤ 2,500 sq ft) | 1× RT-BE58 Go primary only — no extension needed |
| Multi-room EOC or large shelter (2,500–7,500 sq ft) | 1× RT-BE58 Go primary + 1 AiMesh node (wireless or wired backhaul) |

# Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

## Pi 500 vs Pi 500+ vs Pi 400 — Comparison

| SPEC | Pi 400 (2020) | Pi 500 (Dec 2024) ★ Recommended | Pi 500+ (Sept 2025) |
|---|---|---|---|
| Processor | Cortex-A72 @ 1.8 GHz (Pi 4 generation) | **Cortex-A76 @ 2.4 GHz (Pi 5 generation)** | Cortex-A76 @ 2.4 GHz (Pi 5 generation) |
| RAM | 4 GB DDR4 | **8 GB LPDDR4X** | 16 GB LPDDR4X |
| Storage | microSD only | microSD only (32 GB A2 included) | 256 GB NVMe built-in + microSD slot |
| Keyboard | Membrane | Membrane | Mechanical (Gateron Blue KS-33) |
| Wi-Fi | 802.11ac dual-band | 802.11ac dual-band | 802.11ac dual-band |
| Ethernet | Gigabit | Gigabit | Gigabit |
| USB ports | 2× USB 3.0, 1× USB 2.0 | 2× USB 3.0, 1× USB 2.0 | 2× USB 3.0, 1× USB 2.0 |
| Display output | 2× micro-HDMI | 2× micro-HDMI | 2× micro-HDMI |
| Production until | Jan 2028 | **TBD (current model)** | TBD (current model) |
| Best for FieldComms? | Acceptable — older chip, only 4 GB RAM | **✓ YES — Pi 5 chip, 8 GB RAM, current generation** | Overkill for browser use Justified only if also running heavy Linux apps locally |

## Raspberry Pi Monitor — 15.6"

| SPEC | DETAIL |
|---|---|
| Panel | 15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz |
| Video input | Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500 |
| Audio | 2× front-facing stereo speakers, 3.5mm headphone jack |
| Power | USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume |
| Mounting | Integrated angle-adjustable kickstand, VESA mount, wall hanger |
| Dimensions | Slim 15.6" form — similar footprint to a laptop |
| Colors | Red/White (matching Pi 500 keyboard) or Black |
| Where to buy | raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon |

■ The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit ($120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

## Operator Workstation — Bill of Materials (Per Station)

| ITEM | MODEL / SPEC | WHERE TO BUY |
|---|---|---|
| **— Standard Workstation (Pi 500) —** | | |
| Raspberry Pi 500 keyboard computer | Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard | raspberrypi.com Adafruit · Amazon |
| Raspberry Pi Monitor | 15.6" Full HD IPS, built-in speakers, kickstand, VESA mount | raspberrypi.com GigaParts · Amazon |
| micro-HDMI to HDMI cable | 1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit ($120) — verify before ordering separately | raspberrypi.com Amazon |
| USB mouse | Any USB or Bluetooth mouse — included in Pi 500 Kit | Amazon |
| USB-C power supply for Pi 500 | Official Raspberry Pi 27W USB-C PD power supply (or equivalent) | raspberrypi.com Amazon |
| USB-C power supply for monitor (optional, for full brightness) | 5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors) | Amazon |
| **— Premium Workstation (Pi 500+) —** | | |
| Raspberry Pi 500+ keyboard computer | Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB | raspberrypi.com Adafruit · Amazon |
| Raspberry Pi Monitor + cables + mouse | Same as Standard Workstation above | Same as above |

■ **501(c)(3) Grant Funding Note:** MCESV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

## Recommended Deployment by Position

| POSITION | RECOMMENDED DEVICE | REASON |
|---|---|---|
| Net Control Operator (fixed EOC desk) | **Pi 500 + Pi Monitor** | Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly |
| ICS Section Chief (Planning, Ops, Logistics) | Pi 500 + Pi Monitor or Windows/Mac laptop | Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages |
| Served Agency Staff (EOC or shelter) | Any tablet, Chromebook, or Pi 500 | Observer Mode and ICS forms work on any browser — no special hardware needed |
| Winlink / JS8Call Operator | Windows laptop (required) | Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively |

| | | |
|---|---|---|
| Mobile / Field Operator | Phone or tablet | FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed |

■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.

| | |
|---|---|
| Large building or campus (7,500–20,000 sq ft) | 1× primary + 2–3 nodes, wired backhaul recommended for all nodes |
| Outdoor SAR staging area | 1× primary at command post + RT-BE58 Go nodes at field positions on battery |

> ■ You do not need to pre-purchase AiMesh nodes for every deployment. Start with the primary RT-BE58 Go and add nodes only when coverage is insufficient. Pairing a new node takes under 5 minutes on site.

# 3. Before You Begin — Prerequisites

Complete these prerequisites before running the installer. The installer requires an internet connection for package downloads, the FCC database, and Kiwix ZIM files. After installation, the server operates fully offline.

## 3.1 Flash the Operating System

Install **Raspberry Pi OS Lite (64-bit)** or **Raspberry Pi OS Desktop (64-bit)** using the Raspberry Pi Imager tool. Both work well on the Pi 5 with 16 GB RAM. Ubuntu Server 24.04 LTS is also supported.

| OS EDITION | PROS | CONS |
|---|---|---|
| Raspberry Pi OS Lite (64-bit) **Recommended for production** | Minimal overhead, faster boot, all RAM available to FieldComms services, smaller attack surface | No local browser or GUI — all interaction via SSH or EMCOMM-NET browser |
| Raspberry Pi OS Desktop (64-bit) *Good for setup and troubleshooting* | GUI for initial Wi-Fi setup, Raspberry Pi Configuration tool, local browser for testing, visual feedback during troubleshooting — very useful if you are newer to Linux | ~200–300 MB additional RAM used by desktop compositor — negligible on 16 GB Pi 5 |

■ In Raspberry Pi Imager: click the ■ gear/settings icon before writing. Set your hostname (e.g. fieldcomms), enable SSH, set a username and password (e.g. fieldcomms / your-password), and optionally configure your Wi-Fi credentials for initial internet access.

## 3.2 First Boot — Update the System

First boot procedure differs slightly depending on which OS edition you chose:

| EDITION | FIRST BOOT PROCEDURE |
|---|---|
| Raspberry Pi OS Lite (headless) | **SSH into the Pi** from another computer on the same network: `ssh fieldcomms@[pi-ip-address]` Find the Pi's IP in your router's DHCP client list, or use `ping fieldcomms.local` if mDNS is available. Then run the update commands in the code block below. |
| Raspberry Pi OS Desktop | **Connect a monitor, keyboard, and mouse** to the Pi. The desktop loads automatically. Use the **Raspberry Pi Configuration** tool (Preferences menu) for any initial setup. Open a **Terminal** window and run the update commands below. You can also SSH in if you prefer. |

Run a full system update before installing FieldComms (works the same on both editions — Terminal or SSH):

```
# Update all packages to latest versions
sudo apt-get update && sudo apt-get full-upgrade -y

# Reboot to apply kernel updates
sudo reboot
```

## 3.3 Verify Disk Space

```
# Check available space — should show 20GB+ free for Tier 2 install
df -h /

# Verify RAM
free -h
```

## 3.4 Internet Connection

Connect the Pi to the internet via Ethernet before running the installer. WiFi can be used, but Ethernet is preferred for downloading large files (FCC database ~600 MB, Kiwix ZIMs up to 25 GB).

■ Wi-Fi is provided by the ASUS RT-BE58 Go router — the Pi does NOT run hostapd. During installation, connect the Pi to the internet via Ethernet through the UniFi switch and ASUS router. After installation, the Pi is always reachable at 192.168.50.1 via wired Ethernet. Configure the ASUS router first (Step 3) before testing FieldComms.

1 NETWORK HARDWARE SETUP — ASUS ROUTER + UNIFI SWITCH

# 4. Step 1: Network Hardware Setup

Complete this step before running the FieldComms installer. The ASUS router must be configured with the correct LAN subnet (192.168.50.x) before FieldComms is reachable at http://192.168.50.1.

## 1.1 Physical Wiring

```
ASUS RT-BE58 Go WAN port  →  Ethernet uplink (EOC internet) or leave unplugged for offline
ASUS RT-BE58 Go LAN port  →  UniFi Switch Flex 2.5G-5, Port 1 (uplink)
UniFi Switch Port 2       →  Raspberry Pi 5 Ethernet
UniFi Switch Port 3       →  Windows laptop (optional, or use Wi-Fi)
UniFi Switch Ports 4–5    →  Spare (second TNC, Starlink, etc.)

Power:
  ASUS router   — USB-C 18W PD adapter or power bank
  UniFi switch  — included USB-C 5V/1A adapter (or PoE from uplink)
  Pi 5          — 27W USB-C PD supply via Pironman MAX 5 enclosure PSU
```

## 1.2 Configure the ASUS RT-BE58 Go

Power on the router. Connect a laptop to the ASUS default Wi-Fi SSID (printed on the label) or via Ethernet. Open a browser to **http://192.168.1.1** and complete the Quick Setup wizard.

### Critical settings to change:

| SETTING | WHERE | VALUE TO SET |
|---|---|---|
| LAN IP address | `LAN → LAN IP` | 192.168.50.1 (router will reboot — reconnect to http://192.168.50.1) |
| DHCP pool | `LAN → DHCP Server` | Start: 192.168.50.100 End: 192.168.50.200 |
| Wi-Fi SSID (2.4 GHz) | `Wireless → General` | EMCOMM-NET |
| Wi-Fi SSID (5 GHz) | `Wireless → General` | EMCOMM-NET (same SSID, devices auto-select the best band) |
| Wi-Fi password | `Wireless → General` | Set a strong WPA3/WPA2 password and record it on the FIELDCOMMS USB drive |
| Pi DHCP reservation (recommended) | `LAN → DHCP → Manually Assigned IP` | Enter Pi MAC address → assign 192.168.50.1 (or set static IP on Pi — see Step 4) |

■ If the ASUS router is factory-reset in the field, it will revert to 192.168.1.x and FieldComms will be unreachable until you reconfigure it. Write the admin password on a label inside the equipment case.

## 1.3 WAN / Internet Options

The system works fully offline — WAN is optional. Configure it if internet is available at the site.

| WAN SOURCE | SETTING | WHEN TO USE |
|---|---|---|
| No WAN (offline only) | Leave WAN unplugged | Default field posture — all FieldComms tools work without internet |
| Ethernet uplink | Plug site Ethernet into WAN port, set WAN → Automatic IP | EOC with site network — enables NWS alerts, HF prop data, APRS-IS |
| USB smartphone tether | Enable tethering on phone, plug into ASUS USB port, set WAN → USB | Field site with cellular coverage |
| WISP (connect to site Wi-Fi) | ASUS web UI → Operation Mode → WISP | Hospital / shelter with guest Wi-Fi available |

## 1.4 UniFi Switch Flex 2.5G-5

The switch requires no software configuration for basic FieldComms use. Plug in the USB-C power adapter (or connect PoE from the ASUS router if your router provides PoE), connect the cables per Section 1.1, and the switch is ready. Link LEDs on active ports confirm connectivity.

> ■ To manage the switch via the UniFi Network app (VLANs, port stats, firmware updates): install UniFi Network on the EOC laptop, create a local-only site (no cloud account required), and adopt the switch. This is optional — the switch works at full speed without it.

## 1.5 Extending Wi-Fi Coverage — ASUS AiMesh

A single ASUS RT-BE58 Go covers approximately 2,000–2,500 sq ft under typical conditions. For larger deployments — multi-room EOC buildings, large shelters, outdoor staging areas, or multi-floor facilities — Wi-Fi coverage can be extended using **ASUS AiMesh**, which adds additional ASUS routers as wireless or wired nodes on EMCOMM-NET. All nodes share the same SSID (EMCOMM-NET), the same password, and the same 192.168.50.x subnet — devices roam seamlessly between nodes with no reconfiguration required.

### Compatible AiMesh node routers (examples):

| ROUTER | FORM FACTOR | BEST USE |
|---|---|---|
| ASUS RT-BE58 Go (same as primary) | Travel / portable | Identical hardware — simplest pairing. Battery-friendly. Good for field extensions. |
| ASUS ZenWiFi Pro ET12 | Tri-band desktop | Large indoor venues, EOC buildings — very high capacity and range. |
| ASUS RT-AX88U Pro | Desktop | High-density environments, many simultaneous devices. |
| Any AiMesh-capable ASUS router | Varies | Most ASUS routers released after 2018 support AiMesh. Check asus.com/aimesh. |

### Adding an AiMesh Node — Step by Step

The primary router (already configured as EMCOMM-NET) acts as the AiMesh router. Additional units become AiMesh nodes.

| STEP | ACTION |
|---|---|
| 1 | **Factory reset the node router** if it has been previously configured. Hold the reset button for 5–10 seconds until the power LED flashes. |
| 2 | **Power on the node** and place it within range of the primary router (within 30 ft for initial pairing — it can be moved after). |
| 3 | **On the primary router** (http://192.168.50.1) go to: **AiMesh** → **Add AiMesh Node**. The router scans for nearby nodes. |
| 4 | **Select the new node** from the list when it appears. Click **Connect**. The primary router pushes the EMCOMM-NET SSID, password, and 192.168.50.x subnet to the node automatically. This takes 1–3 minutes. |
| 5 | **Move the node** to its final position once pairing completes. The node can connect to the primary via Wi-Fi backhaul (wireless) or Ethernet backhaul (wired — preferred for reliability). |
| 6 | **Test coverage** by walking the area with a phone connected to EMCOMM-NET and confirming http://192.168.50.1 loads throughout the space. |

| CONNECTION TYPE | HOW TO SET UP | WHEN TO USE |
|---|---|---|
| Wireless backhaul (no extra cable) | No setup needed — default after pairing. Node connects to primary via Wi-Fi automatically. | When running cable is not practical — outdoor areas, across hallways, temporary sites. |
| Wired Ethernet backhaul (recommended) | Run a CAT5e/6 cable from any UniFi switch port to the node's LAN port. AiMesh detects the wired connection automatically within 60 seconds. | Permanent or semi-permanent EOC setups. Fastest and most reliable — frees up Wi-Fi spectrum for client devices. |

■ All AiMesh nodes share the EMCOMM-NET SSID and 192.168.50.x subnet. Devices roam automatically between nodes — operators do not need to manually switch Wi-Fi networks when moving around the site. The Pi remains at 192.168.50.1 regardless of which node a device connects through. Up to 8 AiMesh nodes are supported on most ASUS routers.

**2** **DOWNLOAD & RUN THE FIELDCOMMS INSTALLER**

# 5. Step 2: Download & Run the FieldComms Installer

Transfer the FieldComms package to the Raspberry Pi. There are two methods — choose the one that works best for your setup.

### Method A — Direct download to the Pi (requires internet on Pi)

**Lite:** SSH into the Pi and run these commands. **Desktop:** open a Terminal window (taskbar → Terminal) and run the same commands.

```
# Create working directory
mkdir -p ~/fieldcomms-install && cd ~/fieldcomms-install

# Download the FieldComms package
# (replace URL with your actual download link)
wget -O fieldcomms-v1.zip https://your-server/fieldcomms-v1.zip

# Unzip the package
unzip fieldcomms-v1.zip
cd fieldcomms
```

### Method B — Copy from your computer using SCP (Lite/headless)

```
# Run this on YOUR COMPUTER, not the Pi
# Copy the downloaded zip to the Pi
scp fieldcomms-v1.zip fieldcomms@[pi-ip-address]:~/

# Then SSH into the Pi and unzip
ssh fieldcomms@[pi-ip-address]
mkdir -p ~/fieldcomms-install
mv fieldcomms-v1.zip ~/fieldcomms-install/
cd ~/fieldcomms-install && unzip fieldcomms-v1.zip
cd fieldcomms
```

### Method B2 — Download directly on the Pi Desktop (Desktop edition)

If running Raspberry Pi OS Desktop, transfer the zip without using SCP or SSH:

| OPTION | HOW |
|---|---|
| Browser download | Open **Chromium** on the Pi desktop. Go to your download URL or GitHub release page. Download **fieldcomms-v1.zip** — it saves to ~/Downloads/ automatically. |
| USB drive | Copy fieldcomms-v1.zip to a USB drive on your computer. Insert the USB into the Pi. The desktop auto-mounts it — open **File Manager**, drag the zip to your home folder. |
| Samba / network share | If your computer shares a network folder, open **File Manager** → **Go** → **Enter Location** and browse to **smb://[your-computer-ip]/share** to copy the file. |

Once the zip is on the Pi, open a **Terminal** and continue:

```
# If downloaded to ~/Downloads/:
mkdir -p ~/fieldcomms-install
cp ~/Downloads/fieldcomms-v1.zip ~/fieldcomms-install/

# If copied from USB or moved manually:
mkdir -p ~/fieldcomms-install
mv fieldcomms-v1.zip ~/fieldcomms-install/

# Unzip (same for all methods):
cd ~/fieldcomms-install && unzip fieldcomms-v1.zip
cd fieldcomms
```

### Method C — USB drive transfer (fully offline)

**Lite:** commands below via SSH. **Desktop:** use File Manager to copy the zip from USB to your home folder, then open a Terminal for the unzip step.

```
# Copy fieldcomms-v1.zip to a USB drive on your computer
# Insert USB drive into the Pi, then:
sudo mount /dev/sda1 /mnt   # or check lsblk for your device
cp /mnt/fieldcomms-v1.zip ~/fieldcomms-install/
cd ~/fieldcomms-install && unzip fieldcomms-v1.zip
cd fieldcomms
```

✓ Verify the file extracted correctly: ls -la should show directories: html/, python/, scripts/, systemd/, udev/, docs/

3 RUN THE INSTALLER

# 5. Step 2: Run the Installer

Launch the interactive installer with root privileges:

**Lite:** run via SSH. **Desktop:** open a Terminal window from the taskbar or Accessories → Terminal. The installer runs identically on both.

```
cd ~/fieldcomms-install/fieldcomms
sudo bash scripts/install.sh
```

The installer will display the FieldComms banner and run through a series of pre-flight checks before prompting for configuration.

## Installer Pre-Flight Checks

| CHECK | WHAT IT VERIFIES | ACTION IF FAILED |
| --- | --- | --- |
| Operating System | Detects Raspbian, Ubuntu, or Debian | Warning shown — install continues on unsupported OS |
| RAM | Minimum 1 GB RAM detected | Warning shown — 2 GB+ recommended |
| Disk Space | Minimum 4 GB free space | Warning shown — 8 GB+ recommended |
| Architecture | arm64, armv7l, or x86_64 | Kiwix binary download selects correct build |
| Root Privileges | Script is running as root (sudo) | Error — must re-run with sudo |

## Installation Profiles

When the installer starts, it asks you to choose a profile:

| PROFILE | DESCRIPTION | USE CASE |
| --- | --- | --- |
| **1 — Full Installation** | All services + Python APIs + WiFi AP configuration + FCC DB + Kiwix | Primary field deployment. Recommended for most users. |
| 2 — Server Only | Python APIs + Web frontend. No WiFi AP configuration. | When another device handles the AP, or using wired LAN only |
| 3 — Web Only | Copies HTML files only. No Python services installed. | Update web pages on an existing installation |
| 4 — Update | Updates files in an existing installation without reconfiguring | Applying FieldComms updates after initial install |

3 INSTALLER CONFIGURATION OPTIONS

# 6. Step 3: Installer Configuration Options

After selecting a profile, the installer prompts for the following configuration values. All have defaults shown in brackets.

| PROMPT | DEFAULT | DESCRIPTION |
|---|---|---|
| `Station callsign` | `W8ABC` | Your amateur callsign. Patched into all HTML pages and the AP beacon. |
| `Station latitude` | `42.3153` | Decimal degrees. Used for APRS station marker, NWS alerts, propagation, distance calc. |
| `Station longitude` | `-88.4473` | Decimal degrees (negative = West). Default: Woodstock IL (McHenry County seat). |
| `WiFi AP SSID` | `EMCOMM-NET` | The WiFi network name that field devices connect to. |
| `WiFi AP password` | `fieldcomms2026` | WPA2 password for EMCOMM-NET. Change this for operational security. |
| `Server IP address` | `192.168.50.1` | The Pi's static IP on the EMCOMM-NET WiFi. Devices browse to http://192.168.50.1/ |
| `Download FCC database` | `N` | ~600 MB download of the FCC amateur license database. Enables offline callsign lookup. |
| `Kiwix content tier` | `1` | Offline library: 0=none, 1=essential (~2.5 GB), 2=extended (~10 GB), 3=full (~25 GB). |

## Example Configuration Session

```
Select installation profile [1-4, default=1]: 1

Station callsign (e.g. W8ABC): W8OAN
Station latitude [42.3153]: 42.3153
Station longitude [-88.4473]: -88.4473
WiFi AP SSID [EMCOMM-NET]: EMCOMM-NET
WiFi AP password [fieldcomms2026]: ARES2026!
Server IP address [192.168.50.1]: 192.168.50.1
Download FCC amateur database? (~600MB) [y/N]: y
Kiwix content tier [0-3, default=1]: 2

Configuration summary:
  Profile:    1
  Callsign:   W8OAN
  WiFi SSID:  EMCOMM-NET
  Server IP:  192.168.50.1
  Kiwix tier: Tier 2

Proceed with installation? [Y/n]: Y
```

## What the Installer Does Automatically

- Installs system packages: Python 3, nginx, kiwix-tools, rsync, gpsd, ufw, mdadm, java
- Creates the **fieldcomms** system user and directory structure at /opt/fieldcomms/
- Creates a Python virtual environment and installs Flask, flask-cors, requests, gpsd-py3
- Deploys all 30 HTML pages to /opt/fieldcomms/html/ and patches your callsign and coordinates into each file
- Installs all 14 systemd service and timer units and enables them at boot
- Configures nginx with the correct web root (/opt/fieldcomms/html/) and proxy rules for all API ports (5050, 5051, 5055, 5056, 8080–8090)
- **Does NOT configure a Wi-Fi access point** — Wi-Fi is handled by the ASUS RT-BE58 Go router. The Pi connects as a wired client only.
- Configures the Pi static IP (192.168.50.1/24) on the Ethernet interface (eth0) via NetworkManager
- Configures the firewall (ufw) — opens all 11 required ports
- Downloads and installs YAAC (Java APRS client) and Graywolf to /opt/yaac/ and /opt/graywolf/
- Installs the USB backup udev rule (plug in a USB drive labeled FIELDCOMMS to trigger auto-backup)
- Downloads and installs Pat Winlink (browser-based Winlink backup client, port 8090)
- Runs the Kiwix ZIM downloader for the selected content tier
- Downloads the FEMA ICS forms PDFs to /opt/fieldcomms/data/ics_forms/ (22 forms)
- Optionally downloads and builds the FCC amateur license database (~600 MB)
- Runs the apply_theme.py check to verify HTML theme consistency
- Starts all services and verifies they are running

■ The ASUS RT-BE58 Go router must be configured before running the installer — see Step 1. The Pi does NOT run hostapd or dnsmasq. All Wi-Fi, DHCP, and SSID management is handled by the ASUS router.

**4 RASPBERRY PI 5 — STATIC IP CONFIGURATION**

# 7. Step 4: Raspberry Pi 5 — Static IP Configuration

The Pi must always be reachable at **192.168.50.1**. If the installer already set this via a DHCP reservation in the ASUS router, verify with ip addr show eth0 and skip to Step 5 if 192.168.50.1/24 is shown. Otherwise follow the path below that matches your OS edition.

## Path A — Raspberry Pi OS Lite (SSH)

Connect to the Pi over SSH from any computer on the same network:

```
# SSH into the Pi (use the IP shown in your router DHCP table)
ssh fieldcomms@[current-pi-ip]

# Find the Ethernet connection name
nmcli con show
# Usually: "Wired connection 1" or "Preconfigured Connection 1"

# Set static IP (replace name if different)
sudo nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.50.1/24 \
  ipv4.method manual

# Apply the change
sudo nmcli con up "Wired connection 1"

# Verify — should show inet 192.168.50.1/24
ip addr show eth0

# Test FieldComms is reachable
curl -s http://192.168.50.1/ | head -3
```

■ After setting the static IP, reconnect your SSH session using ssh fieldcomms@192.168.50.1 for all future access.

## Path B — Raspberry Pi OS Desktop (GUI)

With a monitor and keyboard attached, use the desktop network manager to set the static IP — no terminal required:

| STEP | ACTION |
|---|---|
| 1 | Click the **network icon** in the taskbar (top-right corner). It looks like two arrows or a Wi-Fi symbol depending on connection type. |
| 2 | Click **Advanced Options** → **Edit Connections**. The Network Connections window opens. |

| | |
|---|---|
| 3 | Select your **Wired connection** (usually "Wired connection 1") and click the ■ **gear / edit** icon. |
| 4 | Click the **IPv4 Settings** tab. |
| 5 | Change **Method** from "Automatic (DHCP)" to **Manual**. |
| 6 | Click **Add** under the Addresses section and enter: Address: 192.168.50.1 Netmask: 255.255.255.0 Gateway: (leave blank) |
| 7 | Click **Save**, then close the Network Connections window. |
| 8 | Disconnect and reconnect the Ethernet cable, or reboot the Pi, to apply the new static IP. |
| 9 | Open the **Terminal** and verify: ip addr show eth0 You should see **inet 192.168.50.1/24** in the output. |
| 10 | Open the **Chromium browser** on the Pi and go to **http://192.168.50.1** to confirm FieldComms loads locally. |

■ Alternatively, open a Terminal on the desktop and use the same nmcli commands shown in Path A — the terminal method works identically on the Desktop edition. Some users find the terminal faster even on the Desktop version.

## Path C — Raspberry Pi Configuration Tool (Desktop)

The Raspberry Pi Configuration tool provides a simpler GUI for network settings on some OS versions:

```
# Open from the desktop menu:
Preferences → Raspberry Pi Configuration → Localisation tab

# Or open a Terminal and launch it directly:
sudo raspi-config
# Navigate to: System Options → Network → (set hostname if needed)

# For static IP via raspi-config:
# Network Options → (not always available in newer OS versions)
# If not available, use the nmcli method in Path A instead
```

■ Static IP configuration via raspi-config was removed in Raspberry Pi OS Bookworm. If raspi-config does not show a network IP option, use Path A (nmcli terminal) or Path B (GUI Network Connections) instead.

# 7b. RAID 1 NVMe Setup — Pironman MAX 5

The Pironman MAX 5 enclosure has two M.2 NVMe slots. Configure them as a RAID 1 mirror so both drives contain identical data. If one drive fails, the system continues running on the surviving drive with no data loss and no operator action required.

> ■ Complete this step before running the FieldComms installer. The RAID array must be assembled and the OS installed on it before FieldComms is deployed. If your Pi OS is already on a single drive, back it up first.

## Physical Installation

```
1. Install both 1 TB NVMe SSDs into the Pironman MAX 5 M.2 slots.
   Slot 1 (primary): will become /dev/nvme0n1
   Slot 2 (mirror):  will become /dev/nvme1n1

2. Boot from a Raspberry Pi OS microSD card for the initial RAID setup.
   (After RAID is built and OS is installed on the array, the SD card is removed.)
```

## Build the RAID 1 Array

```
# Install mdadm (RAID management tool)
sudo apt-get install -y mdadm

# Verify both NVMe drives are detected
lsblk | grep nvme
# Should show: nvme0n1 and nvme1n1

# Create the RAID 1 array
# md0 = the array device name
sudo mdadm --create /dev/md0 \
  --level=1 \
  --raid-devices=2 \
  /dev/nvme0n1 /dev/nvme1n1

# Type "yes" when prompted to confirm

# Monitor build progress (initial sync takes ~30 min for 1 TB)
watch -n 5 cat /proc/mdstat
# Wait until [UU] appears — both drives are in sync
```

## Install the OS on the RAID Array

```
# Create a filesystem on the RAID array
sudo mkfs.ext4 -L fieldcomms-raid /dev/md0

# Mount it
sudo mkdir -p /mnt/raid
sudo mount /dev/md0 /mnt/raid

# Option A: Flash Raspberry Pi OS directly to /dev/md0 using rpi-imager
#           (select /dev/md0 as the target — requires rpi-imager on Linux)

# Option B: Copy running OS to RAID array (if booted from SD card)
sudo apt-get install -y rsync
sudo rsync -axv --progress / /mnt/raid/

# Update /etc/fstab on the RAID array to mount from /dev/md0
sudo sed -i 's|PARTUUID=.*/ |/dev/md0 / |' /mnt/raid/etc/fstab

# Save RAID configuration so it persists across reboots
sudo mdadm --detail --scan | sudo tee -a /mnt/raid/etc/mdadm/mdadm.conf
sudo update-initramfs -u -k all
```

## Boot from RAID and Verify

```
# Update the bootloader to point to /dev/md0
# Edit /boot/cmdline.txt and change root= to:
sudo nano /boot/cmdline.txt
# Change: root=PARTUUID=xxxx  to:  root=/dev/md0

# Reboot — remove the SD card after shutdown, before power-on
sudo reboot

# After booting from RAID, verify:
cat /proc/mdstat
# Expected output:
# md0 : active raid1 nvme1n1[1] nvme0n1[0]
#       976762880 blocks super 1.2 [2/2] [UU]

# [UU] = both drives healthy and in sync
# [_U] = first drive failed (system still running on second)
# [U_] = second drive failed (system still running on first)

# Check RAID status anytime:
sudo mdadm --detail /dev/md0

# The FieldComms health monitor will report RAID status
# at http://192.168.50.1/health.html
```

## RAID Monitoring & Drive Replacement

| SCENARIO | WHAT HAPPENS | RECOVERY PROCEDURE |
|---|---|---|

| | | |
|---|---|---|
| One drive fails | System continues running on the surviving drive. /proc/mdstat shows [_U] or [U_]. | 1. Power down. 2. Remove failed drive. 3. Insert new 1 TB NVMe. 4. Power on. 5. Run: sudo mdadm /dev/md0 --add /dev/nvme1n1. 6. Array rebuilds automatically (~30 min). |
| Check RAID health | Routine monitoring — no issue. | cat /proc/mdstat or sudo mdadm --detail /dev/md0 |
| Both drives show [UU] | Normal healthy state — both drives in sync. | No action needed. |

■ RAID 1 protects against drive failure — it is NOT a backup. Both drives contain identical data, so accidental deletion or corruption affects both drives simultaneously. Continue to use the USB auto-backup (insert FIELDCOMMS USB drive) for regular data backups.

5 KIWIX OFFLINE LIBRARY SETUP

# 8. Step 5: Kiwix Offline Library Setup

Kiwix provides an offline web library served on port 8081. All devices on EMCOMM-NET can browse it at http://192.168.50.1:8081. Content is packaged as ZIM files — compressed offline archives of websites including Wikipedia, WikiMed, iFixit, and more.

## Available Content Tiers

| TIER | NAME | SIZE | DESCRIPTION |
|---|---|---|---|
| **1 — Essential** | WikiMed — Medical Encyclopedia | ~471 MB | Offline medical encyclopedia: symptoms, treatments, drugs, procedures. Critical for mass-casualty events. |
| | Wikipedia (Mini) | ~1.2 GB | Compact English Wikipedia — essential facts, geography, science. |
| | Wikivoyage | ~820 MB | Emergency shelters, evacuation routes, local resources and travel logistics. |
| **2 — Extended** | Wikibooks — How-To & Field Manuals | ~4.2 GB | First aid, survival, ham radio, electronics, cooking, field skills. |
| | iFixit — Equipment Repair Manuals | ~3.1 GB | Step-by-step repair guides for radios, computers, generators, medical equipment. |
| **3 — Full Suite** | Wikipedia Medicine — Full Medical | ~8.5 GB | Complete clinical reference: diseases, drugs, anatomy, surgery, public health. |
| | Wikiversity — Training & Education | ~3.4 GB | ICS training, CERT, ham radio courses, first responder education. |
| | Electronics Stack Exchange | ~2.2 GB | Q&A for antenna design, radio circuits, power systems, field equipment repair. |

## Managing Kiwix After Installation

```
# Check what is installed and service status
sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh --status

# Add Tier 2 content later (resumes interrupted downloads)
sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh --tier 2

# List all available ZIM files with sizes
sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh --list

# Check for newer ZIM versions
sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh --update

# Service management
sudo systemctl status kiwix
sudo systemctl restart kiwix    # after adding new ZIM files
sudo journalctl -fu kiwix       # live logs
```

✓ ZIM downloads are resumable. If a download is interrupted (power loss, network drop), simply re-run kiwix_setup.sh with the same --tier flag. curl will resume from where it stopped — no need to start over.

6 FCC AMATEUR DATABASE

# 9. Step 6: FCC Amateur Database

The FCC database provides offline callsign lookup for the entire US amateur license database (~800,000 licensees). Once built, lookups are instant and completely offline. The net control logger uses it to auto-fill operator information when a callsign is entered.

## Build the Database

```
# If you selected "y" during install, it already ran.
# To build or rebuild manually:

sudo -u fieldcomms \
  /opt/fieldcomms/venv/bin/python \
  /opt/fieldcomms/python/build_fcc_db.py

# Expected output:
# Downloading FCC amateur database...
# Processing records... (800k+ records)
# Building SQLite index...
# Done: /opt/fieldcomms/data/fcc.db (approx 600 MB)
```

## Automatic Weekly Refresh

A systemd timer (fcc-refresh.timer) automatically rebuilds the database every Sunday at 03:00 when internet is available. Check its status:

```
sudo systemctl status fcc-refresh.timer
sudo systemctl list-timers fcc-refresh.timer
```

7 APRS SETUP — GRAYWOLF + YAAC

# 10. Step 7: APRS Setup (Graywolf + YAAC)

FieldComms supports two simultaneous APRS clients: **Graywolf TNC** (port 8080) and **YAAC — Yet Another APRS Client** (port 8082). The tactical map merges data from both sources, deduplicates by callsign, and maintains live WebSocket connections to each for real-time updates. Both are *optional* — FieldComms runs fully without them if APRS is not needed.

> The FieldComms installer attempts to download and install both YAAC and Graywolf automatically during Step 2. If internet was available during installation, Java, YAAC, and Graywolf should already be present. Skip to Section 9.3 (YAAC port configuration) and Section 9.4 (TNC connection) if the installer succeeded. Run the steps below only if the automatic install failed.

## 9.1 Verify Automatic Installation

```
# Check if Java is installed
java -version

# Check if YAAC was installed
ls -lh /opt/yaac/YAAC.jar

# Check if Graywolf was installed
ls -lh /opt/graywolf/graywolf.jar

# Check service status
sudo systemctl status yaac
sudo systemctl status graywolf
```

## 9.2 Manual Install — If Automatic Install Failed

Run these commands if the installer did not install Java, YAAC, or Graywolf:

```
# Step 1 — Install Java runtime (required by both clients)
sudo apt install -y default-jre
java -version    # confirm: openjdk 17 or similar

# Step 2 — Install YAAC
sudo mkdir -p /opt/yaac
cd /tmp
wget http://www.ka2ddo.org/ka2ddo/YAAC.zip
sudo unzip YAAC.zip "*.jar" -d /opt/yaac/
# Find and rename the jar if needed:
sudo find /opt/yaac -name "*.jar" | head -5
sudo mv /opt/yaac/YAAC*.jar /opt/yaac/YAAC.jar 2>/dev/null || true
sudo chown -R fieldcomms:fieldcomms /opt/yaac

# Step 3 — Install Graywolf
sudo mkdir -p /opt/graywolf
# Download from: https://github.com/vk2tds/graywolf/releases
sudo wget -O /opt/graywolf/graywolf.jar \
  https://github.com/vk2tds/graywolf/releases/latest/download/graywolf.jar
sudo chown -R fieldcomms:fieldcomms /opt/graywolf

# Step 4 — Enable services
sudo systemctl enable --now yaac
sudo systemctl enable --now graywolf
```

## 9.3 YAAC Port Configuration (Required — First Run)

YAAC must be configured to use port **8082** before it can serve the FieldComms tactical map. This requires one initial run with a desktop/monitor attached, or via X11 forwarding over SSH.

```
# Option A — Run YAAC on a monitor-connected Pi or via VNC
java -jar /opt/yaac/YAAC.jar
# In the YAAC GUI: File → Configure → Web Server tab
#   Port: 8082
#   Enable REST API:  ✓ checked
#   Enable WebSocket: ✓ checked
# Click Save, then close YAAC

# Option B — Run YAAC via SSH with X11 forwarding
ssh -X fieldcomms@192.168.50.1
java -jar /opt/yaac/YAAC.jar
# (configure as above, then close)

# After configuration, start as service:
sudo systemctl start yaac
sudo systemctl status yaac
```

## 9.4 Connect a TNC or Radio Interface

Both YAAC and Graywolf need a TNC or radio interface to receive APRS packets. Configure your interface in YAAC (File → Configure → Ports → Add Port):

| INTERFACE TYPE | PORT TYPE IN YAAC | SETTINGS |
|---|---|---|
| USB TNC (Digirig, SignaLink, etc.) | Serial TNC | Select the USB serial port (e.g. /dev/ttyUSB0), baud rate 9600 or 1200 |
| KISS TNC over TCP (Direwolf) | KISS TNC (TCP) | Host: localhost, Port: 8001 (Direwolf default KISS port) |
| AGW packet engine (Direwolf or AGWpe) | AGW | Host: localhost, Port: 8000 (Direwolf default AGW port) |
| APRS-IS internet (if WAN connected) | APRS-IS | Server: noam.aprs2.net:14580, Filter: r/42.31/-88.44/50 |

## 9.5 Verify APRS is Working

```
# Test YAAC REST API (should return JSON list of heard stations)
curl http://localhost:8082/api/stations

# Test Graywolf REST API
curl http://localhost:8080/api/stations

# Check service logs if no stations appear
journalctl -u yaac -n 30
journalctl -u graywolf -n 30
```

| CLIENT | PORT | REST API ENDPOINT | WEBSOCKET |
|---|---|---|---|
| Graywolf TNC | `8080` | `http://localhost:8080/api/stations` | `ws://localhost:8080/ws` |
| YAAC | `8082` | `http://localhost:8082/api/stations` | `ws://localhost:8082/ws/aprs` |

■ APRS clients are marked as not required in the Pre-Flight Checklist — FieldComms runs fully without them. If APRS is not needed for your activation, skip this step entirely. The tactical map will show empty station lists but all other features are unaffected.

**8** **PRINTER SETUP — THREE CONNECTION OPTIONS**

# 11. Step 8: Printer Setup

FieldComms has print buttons on 17 pages — ICS forms, NTS radiograms, net logs, cheat sheets, roster, resource board, and more. All printing uses the browser's standard **window.print()** function, which sends the job to whatever printer the operator's browser can reach. Three printer connection methods are supported:

## Option A — Client Device Prints to Its Own Printer (Simplest)

Each operator's laptop or tablet prints to its own locally connected or already-configured printer. No Pi configuration needed. When an operator clicks a Print button in FieldComms, the browser print dialog opens on their device and they select their own printer.

> ■ This is the simplest option and requires zero configuration on the Pi. It works immediately with no setup. The only requirement is that the operator's device has a printer configured.

## Option B — USB Printer Shared via Pi (CUPS)

A USB printer is connected directly to the Pi. CUPS (Common Unix Printing System) is installed and shares the printer across EMCOMM-NET so any device on the network can print to it. This is installed automatically by the FieldComms installer (skip with **SKIP_CUPS=1** if not needed).

### B.1 Verify CUPS Installed

```
# Check CUPS is running
sudo systemctl status cups

# Open CUPS web admin (from Pi or any EMCOMM-NET device)
# http://192.168.50.1:631

# If CUPS is not installed:
sudo apt install -y cups cups-bsd cups-client printer-driver-gutenprint avahi-daemon
sudo usermod -aG lpadmin fieldcomms
sudo systemctl enable --now cups avahi-daemon
```

### B.2 Add the USB Printer via CUPS Web Admin

| STEP | ACTION |
|---|---|
| 1 | Plug the USB printer into one of the Pi's USB ports (or into the powered USB hub if one is in use). |
| 2 | On any device connected to EMCOMM-NET, open a browser and go to: **http://192.168.50.1:631** |
| 3 | Click **Administration** → **Add Printer**. |

| | |
|---|---|
| 4 | If prompted for credentials, enter the Pi username and password (e.g. **fieldcomms** / your password). |
| 5 | Your USB printer appears in the **Local Printers** list. Select it and click **Continue**. |
| 6 | Enter a **Name** (e.g. **FieldComms-Printer**), a Description, and Location. Check **Share This Printer**. Click **Continue**. |
| 7 | Select the correct printer driver from the list (or click **Select Another Make/Model** if yours is not shown). Most HP, Canon, Epson, and Brother printers are included in printer-driver-gutenprint. Click **Add Printer**. |
| 8 | Set default options (paper size: Letter, print quality) and click **Set Default Options**. |
| 9 | Click **Print Test Page** to confirm the printer is working. |

## B.3 Add the Shared Printer on Operator Devices

Once the printer is shared in CUPS, field devices on EMCOMM-NET can discover and add it automatically via Bonjour/mDNS:

| DEVICE TYPE | HOW TO ADD THE SHARED PRINTER |
|---|---|
| Windows laptop | Settings → Bluetooth & devices → Printers & scanners → Add a printer. Windows discovers the CUPS shared printer automatically on EMCOMM-NET. Select **FieldComms-Printer** when it appears and click Add. |
| Mac / macOS | System Settings → Printers & Scanners → + Add Printer. The shared printer appears under **Default** tab via Bonjour. Select it and click Add. |
| iPad / iPhone | iOS prints via AirPrint. CUPS with Avahi shares the printer as AirPrint-compatible automatically. Open any FieldComms page, tap Share → Print, and select the printer. |
| Android tablet / phone | Install the **CUPS Print** app (free, Google Play) or use the device's built-in print service. Add the printer at IP 192.168.50.1, port 631, protocol IPP. |
| Chromebook | Settings → Advanced → Printing → Printers → Add Printer. Enter: Name=FieldComms-Printer, Address=192.168.50.1, Protocol=IPP, Queue=/printers/FieldComms-Printer. |

■ The CUPS admin interface is at http://192.168.50.1:631 from any device on EMCOMM-NET. You can add printers, check print queues, cancel jobs, and view error logs from any browser without SSH.

## B.4 Recommended Printers for Field Use

Any USB printer with Linux drivers works. The following are well-tested in field EmComm environments:

| PRINTER | TYPE | WHY IT WORKS WELL |
|---|---|---|
| HP LaserJet P1102w | Laser, USB + Wi-Fi | Excellent Linux driver support via HPLIP. Fast, reliable, low cost per page. Can also connect directly to EMCOMM-NET Wi-Fi (see Option C). |
| Brother HL-L2350DW | Laser, USB + Wi-Fi | Full Linux support. Compact, fast, duplex capable. Good field durability. |
| Canon PIXMA TR150 | Inkjet, USB + Wi-Fi, battery option | Portable battery-powered option for truly mobile deployments. Prints on letter and smaller. Battery lasts ~200 pages. |
| HP OfficeJet 200 | Inkjet, USB + Wi-Fi, battery option | Portable battery printer with larger paper tray. Good field option when shore power is not available. |

# Option C — Network Printer on EMCOMM-NET (Simplest Shared Setup)

A Wi-Fi or Ethernet printer is connected directly to the ASUS router or UniFi switch — no Pi involvement at all. All devices on EMCOMM-NET can discover and print to it without any server configuration.

## C.1 Connect the Printer to EMCOMM-NET

| CONNECTION METHOD | HOW TO SET UP |
| --- | --- |
| Wi-Fi (wireless) | Use the printer's own control panel or setup app to connect it to **EMCOMM-NET**. Enter the EMCOMM-NET password when prompted. The printer receives an IP address in the 192.168.50.100–200 range from the ASUS router DHCP server. |
| Ethernet (wired) | Run a CAT 6 cable from the printer's Ethernet port to any spare port on the UniFi Switch Flex (Ports 3–5). The printer receives an IP address automatically. Check the ASUS router DHCP table to find the printer's assigned IP. |

## C.2 Reserve a Static IP for the Printer (Recommended)

Assign the printer a fixed IP so it never changes between activations:

```
# On the ASUS router web UI (http://192.168.50.1):
LAN → DHCP Server → Manually Assigned IP

# Enter the printer MAC address (printed on a label on the printer)
# Assign a fixed IP in the static range, e.g.: 192.168.50.10
# Click + Add and then Apply

# Reboot the printer to pick up the reservation
# Confirm: ping 192.168.50.10
```

## C.3 Add the Network Printer on Operator Devices

Most modern printers advertise themselves via Bonjour/mDNS and appear automatically in the print dialog on Windows, macOS, and iOS. If the printer does not appear automatically:

```
# Windows — add by IP address:
Settings → Printers & scanners → Add a printer →
"The printer I want isn't listed" →
"Add a printer using TCP/IP address" →
Enter: 192.168.50.10 (or the printer's assigned IP)

# macOS — add by IP address:
System Settings → Printers & Scanners → + →
Click IP tab → Address: 192.168.50.10
Protocol: IPP or Line Printer Daemon

# Android — use manufacturer app or CUPS Print app:
Add printer at IP: 192.168.50.10, port 9100 (RAW) or 631 (IPP)
```

# Printer Option Comparison

| | Option A Client Printer | Option B USB via Pi / CUPS | Option C Network Printer |
|---|---|---|---|
| Pi configuration needed | None | CUPS (auto-installed) | None |
| One printer for all operators | ✗ Each device needs own printer | ✓ Shared across EMCOMM-NET | ✓ Shared across EMCOMM-NET |
| Works without internet | ✓ | ✓ | ✓ |
| iOS / AirPrint support | Requires AirPrint printer | ✓ via CUPS + Avahi | ✓ if AirPrint printer |
| Battery printer support | ✓ | ✓ USB connected | ✓ Wi-Fi connected |
| Best for | Single operator or tablet with own printer | Any USB printer, shared to all devices | Wi-Fi/Ethernet printer already on site |

■ For most K9ESV field activations, <b>Option B (USB printer via CUPS)</b> is recommended — it gives all operators on EMCOMM-NET access to a single shared printer with no per-device setup beyond adding the printer once. A Brother HL-L2350DW or HP LaserJet plugged into the Pi USB hub covers all ICS forms, net logs, and cheat sheets needed for a typical activation.

9 PAT WINLINK — VERIFY & CONFIGURE

# 12. Step 9: Pat Winlink — Verify & Configure

Pat Winlink is installed automatically by the FieldComms installer. It runs as pat.service and provides a browser-based backup Winlink client at http://192.168.50.1:8090. This step verifies it is running and adds your Winlink password.

## 8.1 Verify Pat is Running

```
# Check pat.service status
sudo systemctl status pat
# Expected: Active: active (running)

# If not running:
sudo systemctl start pat && sudo systemctl enable pat

# Confirm port 8090 is listening
ss -tlnp | grep 8090
```

## 8.2 Add Your Winlink Password

The installer pre-configures Pat with your callsign. Add your Winlink password:

```
sudo nano /opt/fieldcomms/.config/pat/config.json

# Verify / update these fields:
#   "mycall": "K9ESV"                 ← set by installer
#   "secure_login_password": "xxxxx"   ← add your Winlink password
#   "http_addr": "0.0.0.0:8090"        ← allows EMCOMM-NET access

# Save (Ctrl+O, Enter, Ctrl+X) then restart:
sudo systemctl restart pat

# Open Pat from any EMCOMM-NET device:
# http://192.168.50.1:8090
```

■ Pat is the backup Winlink client on the Pi. Primary Winlink operations use Winlink Express on the Windows laptop with IC-7300 + VARA HF — configured in Step 9.

9 WINDOWS LAPTOP — WINLINK EXPRESS + JS8CALL

# 12. Step 9: Windows Laptop Setup

The Windows laptop is the primary HF digital station. It runs Winlink Express and JS8Call with the IC-7300 connected via a single USB cable. Connect the laptop to EMCOMM-NET (Wi-Fi) or the UniFi switch Port 3 (wired Ethernet).

## 9.1 IC-7300 One-Time Radio Setup

Set these menus on the IC-7300 (required for USB digital mode operation):

| IC-7300 MENU PATH | SETTING |
|---|---|
| `SET → Connectors → CI-V Baud Rate` | 115200 |
| `SET → Connectors → CI-V Address` | 94h (default) |
| `SET → Connectors → CI-V Transceive` | ON |
| `SET → Connectors → USB Send/Keying → USB Send` | RTS |
| `SET → Connectors → MOD Input → USB MOD Level` | 40–50% |
| `COMP button (speech compression)` | OFF |
| `Mode for digital operation` | USB-D |

## 9.2 Install Winlink Express + VARA HF

```
1. Download Winlink Express from: winlink.org/client-software
   Run installer, accept defaults.

2. On first launch:
   • Enter your callsign and Winlink password
   • Enter grid square: EN52 (McHenry County, IL)

3. Settings → Radio Setup:
   • Radio: IC-7300
   • Control Port: (check Device Manager for IC-7300 COM port)
   • Baud Rate: 115200
   • PTT via CAT: checked
   • RTS / DTR: unchecked

4. Settings → Sound Card:
   • Input:  USB Audio CODEC  (IC-7300)
   • Output: USB Audio CODEC  (IC-7300)

5. Download and install VARA HF from: winlink.org → VARA → VARA HF Modem
   In VARA HF Settings → SoundCard:
   • Input:  USB Audio CODEC  (IC-7300)
   • Output: USB Audio CODEC  (IC-7300)

6. Test: Open a VARA HF session in Winlink Express → connect to any
   Winlink RMS gateway on 40m to confirm audio and CAT control work.
```

## 9.3 Install JS8Call

```
1. Download JS8Call from: js8call.com → Windows installer.
   Run installer, accept defaults.

2. Launch JS8Call. Open File → Settings (F2):

   General tab:
   • My Call: K9ESV  (your callsign)
   • My Grid: EN52   (McHenry County Maidenhead grid)

   Audio tab:
   • Input:  USB Audio CODEC  (IC-7300)
   • Output: USB Audio CODEC  (IC-7300)

   Radio tab:
   • Rig: IC-7300
   • PTT Method: CAT
   • Serial Port: (same COM port as Winlink Express)
   • Baud Rate: 115200

   Reporting tab (CRITICAL — enables FieldComms dashboard card):
   • TCP Server Hostname: 0.0.0.0  ← MUST change from 127.0.0.1
   • Enable TCP Server API: checked
   • TCP Server Port: 2442
   • Accept TCP Requests: checked

3. Click OK. Restart JS8Call for API settings to take effect.
4. Set VFO to 7.078 MHz (JS8 40m calling frequency).
5. Mode menu → Normal.
```

## 9.4 Windows Firewall — Allow JS8Call Port 2442

```
1. Search "Windows Defender Firewall with Advanced Security" in Start
2. Click: Inbound Rules → New Rule
3. Select: Port → TCP → Specific port: 2442 → Next
4. Select: Allow the connection → Next
5. Check: Private, Domain → Next
6. Name: JS8Call API → Finish

   Or: when JS8Call first starts, click "Allow access" on the
   Windows security alert popup — rule is created automatically.
```

## 9.5 Configure the JS8Call Dashboard Card

```
# Find the Windows laptop IP on EMCOMM-NET:
# Run in Windows Command Prompt:
ipconfig
# Note the IPv4 Address for the EMCOMM-NET adapter (e.g. 192.168.50.105)

# Set a DHCP reservation (recommended — gives laptop a fixed IP):
# ASUS Web UI → LAN → DHCP Server → Manually Assigned IP
# Enter laptop MAC address → assign 192.168.50.2

# Configure the FieldComms dashboard card:
# 1. Connect to EMCOMM-NET, open http://192.168.50.1
# 2. Find the JS8Call card (purple) in Amateur Radio section
# 3. Tap/click the card
# 4. Enter the Windows laptop IP (e.g. 192.168.50.105)
# 5. Card saves IP and opens http://192.168.50.105:2442
# This is a one-time setup per device on EMCOMM-NET.
```

## 9.6 IC-7300 Sharing Protocol

| SWITCHING TO... | PROCEDURE |
|---|---|
| Winlink Express | Close/disconnect JS8Call → open VARA HF → it reclaims USB audio → open Winlink Express |
| JS8Call | Finish any Winlink TX → close VARA HF → minimize Winlink Express → launch JS8Call → click Connect |

■ OmniRig (free — afreet.com/Products/OmniRig) can share the IC-7300 CAT port between Winlink Express and JS8Call simultaneously. Set both apps to use OmniRig as the CAT interface. Audio switching is still manual.

10 **FIRST BOOT VERIFICATION**

# 13. Step 10: First Boot Verification

After the installer completes (and a reboot if prompted), verify all services are running correctly.

## 11.1 Check Service Status

```
# Check all FieldComms services
sudo systemctl status fcc-lookup health-monitor deadmans ics-platform

# Quick one-line status for all
for svc in fcc-lookup health-monitor deadmans ics-platform kiwix nginx; do
  echo "$svc: $(systemctl is-active $svc)"
done

# Expected output:
# fcc-lookup: active
# health-monitor: active
# deadmans: active
# ics-platform: active
# kiwix: active
# nginx: active
```

## 11.2 Test the Dashboard

```
# From the Pi itself
curl -s http://localhost/ | head -5

# Test API endpoints
curl http://localhost:5050/health      # FCC API health check
curl http://localhost:5051/health      # System health monitor
curl http://localhost:5055/incidents   # ICS platform

# From another device on your network
# Browse to: http://192.168.1.x/   (Pi ethernet IP during setup)
```

## 10.3 Connect to EMCOMM-NET

EMCOMM-NET is provided by the ASUS RT-BE58 Go router, not the Pi. Ensure the router is powered on and configured per Step 1.

✓ From any phone, tablet, or laptop: connect to Wi-Fi network EMCOMM-NET, enter your password, then open a browser to http://192.168.50.1 — the FieldComms dashboard should load.

```
# Verify Pi has the correct static IP
ip addr show eth0
# Should show: inet 192.168.50.1/24

# If not showing static IP, set it:
sudo nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.50.1/24 \
  ipv4.method manual
sudo nmcli con up "Wired connection 1"

# View devices connected to EMCOMM-NET (via ASUS router ARP)
arp -n | grep 192.168.50

# Verify ASUS router is reachable from Pi
ping -c 3 192.168.50.1
```

# 14. Network Architecture Reference

## Network Topology

Wi-Fi is provided by the ASUS RT-BE58 Go router. The Pi is a wired client with a static IP of 192.168.50.1, connected via the UniFi Switch Flex 2.5G-5. The Pi does not run hostapd or dnsmasq.

| DEVICE | IP ADDRESS | ROLE | CONNECTION |
|---|---|---|---|
| `ASUS RT-BE58 Go` | 192.168.50.1 (LAN gateway) | Wi-Fi AP + DHCP server | WAN: site Ethernet or USB tether (optional) |
| `UniFi Switch Flex 2.5G-5` | N/A (Layer 2 switch) | Wired distribution | Port 1 → ASUS router, Port 2 → Pi, Ports 3–5 spare |
| `Raspberry Pi 5` | 192.168.50.1 (static) | FieldComms application server | Wired Ethernet via UniFi switch |
| `Windows laptop` | 192.168.50.2 (recommended static) | Winlink Express + JS8Call station | Wi-Fi (EMCOMM-NET) or wired via UniFi Port 3 |
| `Field devices (phones, tablets)` | 192.168.50.100–200 (DHCP) | Operator browsers | Wi-Fi (EMCOMM-NET, 2.4 GHz or 5 GHz) |

## Setting Pi Static IP (nmcli)

```
# Find the Ethernet connection name
nmcli con show

# Set static IP (replace "Wired connection 1" with actual name)
sudo nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.50.1/24 \
  ipv4.method manual

# Apply
sudo nmcli con up "Wired connection 1"

# Verify
ip addr show eth0  # should show inet 192.168.50.1/24
```

## ASUS Router Admin Access

```
Router admin URL:    http://192.168.50.1
                     (same IP as Pi — ASUS router is the gateway)

Note: Both the Pi (FieldComms) and the ASUS router share 192.168.50.1
because the router is the actual gateway and the Pi has a static host
address. The Pi serves http://192.168.50.1/ (port 80 via nginx) and
the ASUS router admin is at http://192.168.50.1/ only if accessed
via its own management interface — use the ASUS Router app to avoid
confusion, or assign the router a different LAN IP (e.g. 192.168.50.254)
and update the Pi static IP accordingly.
```

■ Recommended: set the ASUS router LAN IP to 192.168.50.254 and the Pi static IP to 192.168.50.1. This keeps the FieldComms URL clean and the router admin clearly separate at http://192.168.50.254.

# 13. Web Dashboard Reference

| PAGE | URL | DESCRIPTION |
|---|---|---|
| Dashboard | / | Main hub — UTC clock, NWS weather alerts, APRS table, tool cards, DMS status |
| Amateur Net Control | /netcontrol.html | Multi-net logger, FCC autofill, precedence, ICS-309 export, server sync |
| Starcom Net Logger | /starcom.html | Public safety net log with Radio IDs, sc- prefix nets, ICS-309 export |
| Net Observer | /observer.html?net=NETNAME | Read-only net viewer — bookmarkable per net, 15-second auto-refresh |
| Member Roster | /roster.html | Directory with 11 certs, 13 equipment fields, 4 roles, CSV import/export |
| Resource Board | /resources.html | Equipment, vehicle, personnel tracking with status cycling |
| Tactical APRS Map | /tactical.html | Leaflet map — Graywolf + YAAC merged, live WebSocket, APRS symbols, messaging |
| Starcom Resource Map | /resmap.html | Unit positioning map with zone and polygon drawing tools |
| Callsign Lookup | /callsign.html | FCC local database search — ?call=W8ABC for direct URL lookup |
| Dead Man's Switch | /deadmans.html | Per-net inactivity monitor — disarmed / armed / warning / triggered states |
| Pre-Flight Checklist | /preflight.html | GO / CAUTION / NO-GO deployment readiness — 7 sections, ~35 items |
| NTS Radiogram | /nts.html | ARRL-format NTS radiogram generator and traffic log |
| ICS-213 Message | /ics213.html | ICS-213 General Message form with rendered print view |
| ICS-214 Activity Log | /ics214.html | Unit activity log with personnel table and timestamped entries |
| Grid Calculator | /grid.html | Maidenhead bidirectional, 4/6-char, distance and bearing, canvas map |
| Repeater Database | /repeaters.html | RepeaterBook offline CSV/JSON import — filter by ARES, SKYWARN, band, mode |
| Facilities Directory | /facilities.html | EOC, hospital, shelter CRUD with radio frequencies and coordinates |
| Cheat Sheets | /cheatsheets.html | Phonetic alphabet, Q-codes, prowords, band plan, CTCSS/DCS, RST |
| Print Center | /printcenter.html | Unified print hub — all forms, incident cover sheet generator |
| HF Propagation | /propagation.html | SFI, K/A index, band conditions, MUF/LUF, ionosphere, 24-hour chart |
| ICS Platform | /ics/ | Full ICS: Command, Operations, Planning, Logistics, Finance sections |
| Kiwix Library | :8081/ | Offline reference library — WikiMed, Wikipedia, iFixit, Wikibooks and more |

# 14. Service & Port Reference

| PORT | SERVICE | SYSTEMD UNIT | DESCRIPTION |
|---|---|---|---|
| 80 | nginx | `nginx.service` | Web frontend — serves all HTML pages from /var/www/html/ |
| 5050 | FCC Lookup Server | `fcc-lookup.service` | Main API: callsign lookup, net logs, roster, resources, APRS, DMS, ICS forms |
| 5051 | Health Monitor | `health-monitor.service` | System health: CPU temp, memory, disk, GPS, all service status |
| 5055 | ICS Platform | `ics-platform.service` | ICS incident management: incidents, objectives, resources, T-cards |
| 8080 | Graywolf APRS | `(Graywolf service)` | APRS TNC client — REST API + WebSocket live stream |
| 8081 | Kiwix Library | `kiwix.service` | Offline web library — WikiMed, Wikipedia, iFixit, etc. |
| 8082 | YAAC APRS | `yaac.service` | YAAC Java APRS client — REST API + WebSocket, configure to port 8082 |
| 8090 | Pat Winlink | `pat.service` | Winlink email-over-radio — Packet, VARA HF, VARA FM |
| 2442 | JS8Call | `(JS8Call app)` | HF digital keyboard messaging |
| 8300 | VARA HF | `(VARA app)` | VARA HF modem (Windows or Wine) |
| 8310 | VARA FM | `(VARA app)` | VARA FM modem (Windows or Wine) |

## Data Paths

| PATH | CONTENTS |
|---|---|
| `/opt/fieldcomms/data/` | All runtime data — nets, forms, roster, resources, mapstate, repeaters |
| `/opt/fieldcomms/data/fcc.db` | SQLite FCC amateur license database (~600 MB) |
| `/opt/fieldcomms/python/` | All Python service scripts |
| `/opt/fieldcomms/venv/` | Python virtual environment (Flask, flask-cors, requests, gpsd-py3) |
| `/var/www/html/` | All HTML/CSS/JS web pages served by nginx |
| `/opt/kiwix/zim/` | Kiwix ZIM offline library files |
| `/opt/kiwix/library.xml` | Kiwix library index — registered ZIM files |

# 15. Maintenance & Updates

The update.sh maintenance script provides a menu-driven interface for common maintenance tasks:

```
sudo bash /opt/fieldcomms/scripts/update.sh

# Menu options:
#  1) Restart all services
#  2) Stop all services
#  3) Check service status
#  4) View live logs
#  5) Refresh FCC database
#  6) Fetch repeater data (specify state code)
#  7) Update web files from current directory
#  8) Backup data to /tmp
#  9) Show disk usage
```

## USB Auto-Backup

Insert a USB drive formatted with the label **FIELDCOMMS** to trigger an automatic backup of all runtime data. The backup copies /opt/fieldcomms/data/ and /var/www/html/ to a timestamped folder on the USB drive.

```
# Format a USB drive with label FIELDCOMMS (Linux):
sudo mkfs.vfat -n FIELDCOMMS /dev/sda1

# Or on Windows: format as FAT32, set volume label to FIELDCOMMS

# Backup destination on USB:
# /media/fieldcomms/backup/YYYYMMDD_HHMMSS/data/
# /media/fieldcomms/backup/YYYYMMDD_HHMMSS/html/
```

## Scheduled Maintenance

| SCHEDULE | TASK | SYSTEMD UNIT |
|---|---|---|
| Weekly — Sunday 03:00 | Rebuild FCC amateur license database from FCC.gov | `fcc-refresh.timer` |
| On demand | Kiwix ZIM update check and download | `kiwix_setup.sh --update` |
| On USB insert | Auto-backup data to USB drive labeled FIELDCOMMS | `fieldcomms-backup@.service` |

# 16. Troubleshooting

### Issue: Dashboard does not load (http://192.168.50.1 gives no response)

Nginx may not be running or the config may have an error. Run "sudo nginx -t" to check syntax, then "sudo systemctl restart nginx".

```
sudo systemctl status nginx
sudo nginx -t                    # test config
sudo journalctl -fu nginx        # live logs
curl http://localhost/           # test from Pi itself
```

### Issue: Callsign lookup returns no results

The FCC database must be downloaded and built. It takes 5–15 minutes and requires ~600 MB of internet download.

```
# Check if FCC database exists
ls -lh /opt/fieldcomms/data/fcc.db

# Rebuild if missing or corrupted
sudo -u fieldcomms /opt/fieldcomms/venv/bin/python \
  /opt/fieldcomms/python/build_fcc_db.py
```

### Issue: EMCOMM-NET WiFi not visible

EMCOMM-NET is broadcast by the ASUS RT-BE58 Go router, not by the Pi. Check that the router is powered on and the SSID is configured correctly. The Pi does not run hostapd — if you see hostapd errors, they are from a previous install and can be safely disabled.

```
# EMCOMM-NET is provided by the ASUS RT-BE58 Go router, not the Pi

# 1. Verify the ASUS router is powered on (check LED)
# 2. Check the SSID is set to EMCOMM-NET:
#    Open http://192.168.50.254 (or your router LAN IP)
#    Wireless → General → SSID should read EMCOMM-NET

# 3. Verify Pi has correct static IP
ip addr show eth0
# Should show: inet 192.168.50.1/24

# 4. If Pi IP is wrong, reset static IP:
sudo nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.50.1/24 ipv4.method manual
sudo nmcli con up "Wired connection 1"
```

### Issue: Services crash on startup

Python services may fail if the virtual environment is incomplete. Reinstall dependencies as shown. Also verify the data directory exists: sudo mkdir -p /opt/fieldcomms/data/nets /opt/fieldcomms/data/forms

```
# Check which service failed
sudo systemctl status fcc-lookup

# View detailed error log
sudo journalctl -u fcc-lookup --since "10 minutes ago"

# Common fix: reinstall Python dependencies
sudo -u fieldcomms /opt/fieldcomms/venv/bin/pip install \
  flask flask-cors requests gpsd-py3 --upgrade
```

### Issue: Kiwix shows no content at port 8081

Kiwix requires at least one ZIM file to show content. Run kiwix_setup.sh to download ZIM files. After adding new ZIMs, restart kiwix: sudo systemctl restart kiwix

```
sudo systemctl status kiwix
ls -lh /opt/kiwix/zim/

# If ZIM files are present but not loading:
sudo systemctl restart kiwix

# If no ZIM files:
sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh --tier 1
```

### Issue: APRS map shows no stations

The tactical map needs Graywolf or YAAC to be running and receiving APRS packets via a connected TNC. The map will show data when stations are heard on RF. Use the Sources tab in the tactical map to see connection status.

```
# Check Graywolf
curl http://localhost:8080/api/stations

# Check YAAC
curl http://localhost:8082/api/stations

# Both returning empty [] is normal if no RF is heard yet.
# Check TNC is connected and Graywolf/YAAC are configured.
```

# 17. Quick Reference Card

Cut out or print this page for the field.

## ACCESS THE DASHBOARD

**Wi-Fi Network:** EMCOMM-NET (provided by ASUS RT-BE58 Go)

**Dashboard URL:** http://192.168.50.1/

**Pat Winlink:** http://192.168.50.1:8090/

**JS8Call:** http://[windows-laptop-ip]:2442/

**Kiwix Library:** http://192.168.50.1:8081/

**Router Admin:** http://192.168.50.254/ (or your router LAN IP)

## ESSENTIAL COMMANDS

**Restart all services:**

```
sudo bash /opt/fieldcomms/scripts/update.sh
```

**Service status:**

```
sudo systemctl status fcc-lookup
```

**Live logs:**

```
sudo journalctl -fu fcc-lookup
```

**FCC DB rebuild:**

```
sudo -u fieldcomms
/opt/fieldcomms/venv/bin/python
/opt/fieldcomms/python/build_fcc_db.py
```

**Add Kiwix ZIMs:**

```
sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh
--tier 2
```

**Kiwix status:**

```
sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh
--status
```

**USB backup:**

```
Insert USB drive labeled FIELDCOMMS
```

## PORT MAP

**:80** nginx — Web Dashboard (Pi)

**:5050** FCC Lookup / Main API (Pi)

**:5051** Health Monitor (Pi)

**:5055** ICS Platform (Pi)

**:8080** Graywolf APRS (Pi)

**:8081** Kiwix Offline Library (Pi)

**:8082** YAAC APRS (Pi)

**:8090** Pat Winlink — backup (Pi)

**:2442** JS8Call — Windows laptop

**:8300** VARA HF — Windows laptop

**:8310** VARA FM — Windows laptop

## KEY FILE PATHS

**Server home:**

```
/opt/fieldcomms/
```

**Runtime data:**

```
/opt/fieldcomms/data/
```

**FCC database:**

```
/opt/fieldcomms/data/fcc.db
```

**Web pages (nginx):**

```
/var/www/html/
```

**Kiwix ZIM files:**

```
/opt/kiwix/zim/
```

**Install log:**

```
/var/log/fieldcomms-install.log
```

> For support and the latest version, check the FieldComms project documentation. All service logs are available via: sudo journalctl -u SERVICE_NAME Install log: /var/log/fieldcomms-install.log Kiwix setup log: /var/log/fieldcomms-kiwix.log